

2.1 Time evolution equation:

$$w_t = \mu_p + \Psi w_{t-1} + \epsilon_p$$

w_t = state vector at time t

w_{t-1} = state vector at time $(t-1)$

μ_p = mean (bias) of process model (Here = 0)

Ψ → state transition matrix → shows transition from state $(t-1)$ to t .

ϵ_p → Zero mean Gaussian ^{process} noise $\Rightarrow \epsilon_p \sim N(0, \Sigma_p)$

2.2 Measurement Equation:

$$x_t = \mu_m + \Phi w_t + \epsilon_m$$

x_t = measurement vector @ time t .

μ_m = mean (bias) of measurement model (here 0)

Φ → measurement matrix, which maps state to measurement space. In this case, because only posⁿ is observed,

its
$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

w_t → True state at time t

ϵ_m → additive measurement noise $\epsilon_m \sim N(0, \Sigma_m)$

3.2 Analysis of lag size effects:

- Standard KF: ($N=0$)
 - only knows about past & present.
 - susceptible to noise (filter assumes it as "motion" & can thus effect output wrongly)
- Moderate lag: ($N=2-5$)
 - Effectively smoothens high frequency noise rather than blind extrapolation.
- Large lag ($N=10$ or more).
 - Smoothest trajectory by averaging over larger window
 - But, significant computational cost & high delay, thus being unsuitable for real world applications.

4.1 Non-linear components in $g(x)$ and $h(x)$

The system model is

$$g(x_t) = \begin{bmatrix} x_t + \Delta t v_t \cos \theta_t \\ y_t + \Delta t v_t \sin \theta_t \\ \theta_t \\ v_t \end{bmatrix}.$$

The non-linearities arise from the trigonometric terms $\cos \theta_t$ and $\sin \theta_t$, and their coupling with the velocity v_t .

The measurement model is

$$h(x_t) = \begin{bmatrix} x_t \\ y_t \end{bmatrix},$$

which is linear. Hence, $h(x)$ contains no non-linear components.

4.2 Jacobian of $g(x)$

For the state vector $x_t = [x_t, y_t, \theta_t, v_t]^T$, the Jacobian of $g(x)$ is

$$G_t = \begin{bmatrix} 1 & 0 & -\Delta t v_t \sin \theta_t & \Delta t \cos \theta_t \\ 0 & 1 & \Delta t v_t \cos \theta_t & \Delta t \sin \theta_t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

4.3 Jacobian of $h(x)$

The Jacobian of the measurement model is constant:

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}.$$

4.6 Why the Extended Kalman Filter is preferable

The trigonometric dependence on the heading angle θ_t makes the unicycle motion model non-linear. Inaccurate state propagation for this model results from the standard Kalman Filter's assumption of linear system dynamics. By using Jacobians to linearize the non-linear system about the current estimate, the Extended Kalman Filter allows for consistent state and covariance prediction and correction.

4.8 Effect of increasing measurement noise δ_t

The reliability of observations decreases as the measurement noise covariance increases. As a result, the filter depends more on the motion model prediction and the Kalman gain drops. This reduces estimation accuracy by producing smoother estimated trajectories that deviate more from the true states.